LESSON PLAN

BASIC RADIO & AVIONICS (22)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	VIRUS, types of VIRUS, effects of VIRUS, measures to prevent VIRUS attack
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	VIRUS, types of VIRUS, effects of VIRUS, measures to prevent VIRUS attack.
5. Specific Objectives	22.1.1.1 Trainee will be able to explain computer operations and its applications viz hardware, software and networking.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of antivirus software installation and configuration; Safety: Do not run unverified executables; Supervision: Instructor monitors all terminals.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of computer software?
(b) Define the term 'malware' in the context of computer security.
(c) Why is network security critical for naval operations?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Computer Virus • Mechanism of Virus Propagation • Common Virus Types • Effects on Hardware and Software • Prevention Measures	L & Ds	PPT, IB	15 min
(b)	• Analysis of Trojan Horse • Analysis of Worms • Analysis of Ransomware • Impact on Naval Networks • Role of Antivirus Software	L & Ds	PPT, PDF	20 min
(c)	• Implementation of Firewalls • User Awareness Training • Regular System Updates • Data Backup Procedures • Incident Reporting Protocols	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a virus and a worm.
(b) List three specific measures to prevent a virus attack on a naval network.
(c) Describe the potential operational impact of a ransomware attack.

12. Home Assignment and Follow Up	Prepare a brief report on the security protocols required for a computer operating in a Naval Network's environment, referencing the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Security of Data and Information in a Computer Operating in Naval Network’s	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min